

Assignment 1

due date: March 29, 2024

1. Prove that the distribution of $S := \sum_{i=1}^n \xi_i$ is *Gamma*(n, λ), where $\{\xi_i\}$ are i.i.d. exponential random variables with parameter λ .
2. Prove the law of total variance:

$$\text{Var}(X) = \mathbb{E}(\text{Var}(X|Y)) + \text{Var}(\mathbb{E}(X|Y)) .$$

3. The length (in number of characters) of the password for a random account in a particular system is modelled as $L = N + 8$ where N is assumed to following a binomial distribution with mean 2 and variance 1.6. A hacking program can crack a password with length L in T minutes, where

$$E(T|L) = 2L - 4, \quad \text{Var}(T|L) = \frac{L(L-1)}{4} .$$

Determine the mean and variance of T .

4. Let X denote the size of a surgical claim and Y denote the size of the associated hospital claim. An actuary is using a model in which

$$EX = 5, \quad E(X^2) = 27.4, \quad EY = 7, \quad E(Y^2) = 51.4, \quad \text{Var}(X + Y) = 8 .$$

The actuary had assumed that $E(X|Y) = aY + b$. Determine the values of a and b .

5. In a game of odd-man-out, each of three players tosses a coin independently. If the outcomes of their tosses are all the same, then they start over and re-toss their coins. Until the outcomes differ, then the game ends and the player whose outcome differs from those of any other players wins. Suppose that one of the players, Bob, uses a biased coin with probability p of landing head; and the other players use fair coins.
 - (a) Find the probability that Bob will win.
 - (b) Determine the expected number of rounds that they will toss.
6. John von Neumann gave the following procedure to make a fair toss from a biased coin with unknown probability of tossing a Head.
 - (1) Toss the coin twice.
 - (2) If the outcomes are the same, ignore the outcomes and start over from step (1).
 - (3) If the outcomes differ, use the first outcome and ignore the second outcome.Show that this procedure will result in a fair toss.